INTERNSHIP REPORT

FULL STACK DEVELOPMENT

CodeAlpha Virtual Internship Program

Project: ConnectHub — Social Media Platform

Candidate Name:	Kartikey Gupta
Roll Number:	25SCS1003004592
Section / Batch:	2CSE37 / 2025–2029
Programme:	B.Tech Computer Science & Engineering
Host Organization:	CodeAlpha Software Development
Student ID:	CA/DF1/194583
Internship Duration:	10th July 2026 – 10th August 2026
Issue Date:	1st September 2026

1. CANDIDATE'S DECLARATION

I, **Kartikey Gupta** (Roll No: **25SCS1003004592**), student of B.Tech Computer Science and Engineering (Batch 2025–2029, Section 2CSE37), hereby declare that the internship report entitled **"Full Stack Development - ConnectHub Social Media Platform"** is an authentic record of my own work carried out during the Virtual Internship Program at **CodeAlpha** from 10th July 2026 to 10th August 2026.

The information and results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma.

Kartikey Gupta
Roll No: 25SCS1003004592
Department of Computer Science & Engineering
Date: 1st September 2026

2. ACKNOWLEDGEMENT

I express my deep gratitude to **CodeAlpha** for providing me with the opportunity to undertake this virtual internship in Full Stack Web Development [cite: 1, 2, 3]. I extend my sincere thanks to **Swati Sri** (Founder & CEO, CodeAlpha) and the technical mentorship team for their invaluable guidance, encouragement, and continuous support throughout the program duration [cite: 1, 2, 3].

I would also like to express my heartiest gratitude to the Head of Department, faculty coordinators, and staff members of my institute for granting me the permission and encouraging environment to complete this professional development program alongside my academic coursework [cite: 3].

Lastly, I acknowledge my peers and family for their unwavering encouragement and constructive feedback during the architecture and testing phases of the ConnectHub web application project [cite: 2, 3].

3. INTERNSHIP COMPLETION & CREDENTIALS

Verified Credential Details:
• **Student ID:** CA/DF1/194583 [cite: 1, 2, 3]
• **Certificate Title:** Certificate of Virtual Internship – Full Stack Development [cite: 1, 3]
• **Duration:** 10th July 2026 – 10th August 2026 [cite: 1, 2, 3]
• **Issuing Organization:** CodeAlpha Software Development (Recognized by Ministry of Corporate Affairs, Govt. of India) [cite: 1, 2]
• **Date of Issue:** 1st September 2026 [cite: 1, 2, 3]

Letter of Recommendation Summary

As certified by Ms. Swati Sri (CEO & Founder, CodeAlpha) in the official Letter of Recommendation, during the one-month tenure [cite: 2]:

- Exhibited outstanding performance and technical skills in Full Stack Web Development [cite: 2].
- Demonstrated excellent analytical problem-solving abilities and quickly adapted to modern web frameworks and emerging stack technologies [cite: 2].
- Consistently displayed willingness to collaborate, help peers, and deliver full-featured software components under production-like standards [cite: 2].

4. PROJECT DESCRIPTION: CONNECTHUB

4.1 Introduction

In the modern web development landscape, modern social media applications require robust scalable backend systems paired with dynamic, highly responsive user interfaces. **ConnectHub** is a full-stack web application designed and developed during the internship to facilitate real-time content sharing, dynamic user interactions, personalized activity feeds, and secure user management [cite: 3].

4.2 Organization Profile

CodeAlpha is a premier software development and technical training organization focused on empowering aspiring software engineers through immersive virtual internship programs [cite: 1, 2, 3]. Operating under standard corporate compliance (MCA registered, Govt. of India), CodeAlpha offers practical industry-aligned projects across domain tracks including Full Stack Web Development, Data Science, Cyber Security, and Mobile App Development [cite: 1, 2, 3].

4.3 Problem Statement

Building a modern social media platform involves solving key technical hurdles:

1. Securing user credentials and state management using stateless token mechanisms (JWT) with HTTP-only cookies to prevent XSS and CSRF attacks [cite: 3].
2. Designing performant MongoDB database schemas to handle complex relational data (users, posts, likes, comments, follow graphs) in a NoSQL environment [cite: 3].
3. Optimizing query performance for computing personalized news feeds in real-time based on social graph connections [cite: 3].

4.4 Project Objectives

The primary technical objectives accomplished during the project include:

- Architecting RESTful API endpoints for user authentication, content creation, social engagements, and dynamic feeds [cite: 3].
- Implementing secure JWT authentication with password hashing using `bcryptjs` [cite: 3].
- Developing interactive frontend interfaces with modular JavaScript, HTML5, and CSS3 [cite: 3].
- Building dynamic post interaction capabilities: creating, reading, liking/unliking, commenting, and deleting posts [cite: 3].

• Implementing user profile management and follow/unfollow social mechanics [cite: 3].

4.5 Scope of the Project

The scope encompasses end-to-end software development life cycle (SDLC) activities including requirement analysis, database ER design, backend controller construction, API testing using Postman, UI styling, and system integration [cite: 3].

4.6 Technologies and Tools Used

Category	Technologies / Tools	Usage / Description
Frontend	HTML5, CSS3, JavaScript (ES6+)	Responsive UI layouts, DOM manipulation, Async/Fetch API integration [cite: 3].
Backend	Node.js, Express.js framework	RESTful routing, controller design, middleware authentication [cite: 3].
Database	MongoDB, Mongoose ODM	Document modeling, schema validation, population queries [cite: 3].
Security	JSON Web Tokens (JWT), bcryptjs	Stateful/Stateless session tokens, password hashing algorithms [cite: 3].
Tools & VCS	Git, GitHub, VS Code, Postman	Source control, API contract testing, codebase development [cite: 3].

4.7 System Architecture

ConnectHub follows a standard 3-Tier Architecture consisting of:

- **Presentation Tier (Client):** HTML5/CSS3 dynamic interfaces communicating asynchronously with backend services via REST JSON calls [cite: 3].
- **Application Tier (Server):** Express.js HTTP server processing authentication, authorization, validation, and business logic execution [cite: 3].
- **Data Tier (Persistence):** MongoDB cloud database managed via Mongoose schemas storing document collections for Users and Posts [cite: 3].

4.8 Methodology

The project followed an Agile iterative development methodology split into four weekly sprints:

- **Week 1: Setup & Authentication:** Database connection setup, User schema design, register/login endpoints, JWT cookie generation, and password encryption [cite: 3].
- **Week 2: Post Management & Feed:** Post CRUD operations, feed query logic populating author details, and frontend feed view construction [cite: 3].
- **Week 3: Social Interactions:** Like/Unlike mechanism, nesting comment subdocuments/arrays, and dynamic UI state updates [cite: 3].
- **Week 4: Follow System & Finalization:** Follow/Unfollow graph logic, profile endpoints, UI polishing, bug fixing, and documentation [cite: 3].

4.9 Implementation Details

Database Schema Models

The core database entities were structured using Mongoose ODM as follows:

- `User Schema`: Stores username, email, hashed password, bio, avatar, followers array (ObjectId references), and following array [cite: 3].
- `Post Schema`: Stores author (ObjectId ref to User), content text, media URLs, likes array (User references), and comments array containing commenter ID, text, and timestamps [cite: 3].

Key API Endpoints

HTTP Method	Endpoint	Description	Auth Required
`POST`	/api/auth/register	Registers new user account with hashed password	No
`POST`	/api/auth/login	Authenticates credentials and sets JWT cookie	No
`GET`	/api/posts/feed	Fetches posts authored by followed users and self	Yes
`POST`	/api/posts	Creates a new text/media post	Yes
`PUT`	/api/posts/:id/like	Toggles like/unlike status on a target post	Yes
`POST`		Follows or unfollows target user profile	Yes

HTTP Method	Endpoint	Description	Auth Required
	/api/users/:id/follow		

5. LEARNING OUTCOMES & PROFESSIONAL GROWTH

Through the hands-on virtual internship program at CodeAlpha, I acquired significant technical competencies and soft skills [cite: 2, 3]:

5.1 Technical Mastery

- **Full-Stack JavaScript Proficiency:** Mastered asynchronous JS programming (Promises, Async/Await), ES6+ idioms, and REST paradigm design patterns [cite: 3].
- **Secure Backend Development:** Acquired practical experience with JWT token lifecycle management, HTTP-only secure cookies, and password salting with `bcryptjs` [cite: 3].
- **Database Optimization:** Learned how to structure NoSQL collections, leverage Mongoose schema population (`.populate()`), and write optimized aggregation pipelines [cite: 3].

5.2 Software Engineering Practices

- **Clean Architecture:** Applied Separation of Concerns (SoC) principles by separating models, controllers, routes, and custom authentication middleware [cite: 3].
- **Version Control Discipline:** Utilized Git workflow branches, clear commits, and GitHub repository organization standards [cite: 3].
- **API Testing & Debugging:** Experienced systematic endpoint testing using Postman environment variables and handling HTTP error status codes gracefully [cite: 3].

6. CONCLUSION

The Full Stack Development virtual internship at **CodeAlpha** was a transformative learning experience [cite: 1, 2, 3]. Building the **ConnectHub Social Media Platform** allowed me to bridge academic theoretical knowledge with industry-grade practical application [cite: 3].

Over the course of the 1-month program (10th July 2026 to 10th August 2026), I successfully engineered a feature-complete web application encompassing secure authentication, dynamic activity feeds, real-time social engagements, and relational follow systems [cite: 1, 2, 3]. The skills and credentials gained during this program serve as a solid foundation for my future career in software engineering and web application development [cite: 2, 3].

7. BIBLIOGRAPHY / REFERENCES

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